

ORIGINAL PAPER

Alimentador Inteligente para Múltiplos Pets via Internet das Coisas e Visão Computacional

Intelligent Feeder for Multiple Pets via Internet of Things and Computer Vision

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Recebido: yyyy-mm-dd. Revisado: yyyy-mm-dd. Aceito: yyyy-mm-dd.

Abstract

The Abstract (200 words maximum) should be structured to include the following details: **Background**, the context, and purpose of the study; **Results**, the main findings; **Conclusions**, summary, and potential implications. Please minimize the use of abbreviations and do not cite references in the abstract.

Keywords: Keyword1; keyword 2; keyword 3 (Three to five keywords representing the main content of the article, in alphabetic order)

Resumo

O Resumo (com até 200 palavras) deverá ser estruturado para abordar os seguintes detalhes: **Background**, o contexto e propósito do estudo; **Resultados**, os principais encontrados; **Conclusões**, um breve resumo do trabalho e as implicações em potencial. Por favor, evite, na medida do possível, o uso de abreviaturas e não cite referências no resumo.

Palavras-Chave: Palavra-chave 1; palavra-chave 2; palavra-chave 3 (De três a cinco palavras-chave que representem os principais tópicos do artigo, em ordem alfabética)

1 Introduction to this Template

The Revista Brasileira de Computação Aplicada (RBCA) is an **open-access journal** linked to the [Graduate Program in Applied Computing \(PPGCA\)](#) of the [University of Passo Fundo](#), Brazil. RBCA aims to provide to the scientific community article that presents an interdisciplinary perspective of the application of Computing in different areas of knowledge.

This document is the $\text{\LaTeX}$ template for RBCA journal manuscript submissions. **Submissions that do not use the format available in this template will be automatically rejected.**

Articles submitted to RBCA should be between 8 and 15 pages and will be published electronically. The languages accepted by RBCA are English (preferably) and Portuguese. We alert the authors that the preparation of the manuscript should be made carefully, both in its scientific content and in its grammatical correctness.

There are essential commands in the preamble that you will need to modify for your manuscript. Define the paper language (“english” or “brazilian”) in `\documentclass` command and specify your manuscript’s category with the `\papercat{...}` command. See the sample code in the preamble for a sample of how the title, author (names, or-

cid, affiliation, and e-mail) information can be specified. **If the paper is in Portuguese you need to inform the title in English using the `\titleother{...}` command.** Information about this edition of RBCA and publication details of the paper will be filled out by the RBCA's editor.

This template has been edited and validated in the [TexStudio](#)® program and the [Overleaf](#)® Collaborative Writing and Publishing System. Any questions or problems regarding this template report to the RBCA e-mail (rbca@upf.br).

The remainder of this current section will provide some sample $\LaTeX$ code for various elements you may want to include in your manuscript.

2 Manuscript format rules and styles

2.1 Sectional Headings

You can use `\section{...}`, `\subsection{...}` commands to add more sections and subsections to your manuscript. Further sectional levels are provided by `\subsubsection` and `\paragraph`.

2.2 Citations and References

The RBCA use the `alpha-refs` document class option for author-year citations using the `dcu` bibliography style from the [Design Computing Unit](#) of the University of Sydney, a variant of [Harvard style](#). This class is defined in `RBCA_v2.0.cls` class file.

Use the `\citep` or `\cite` commands for citations in the format “(Authors; year)” or “Authors (year)”, respectively. Some examples of citations are shown below:

- Book: [Dongarra et al. \(2003\)](#) and ([Wickham and Grole-mund, 2017](#));
- Book chapter: [Hölbjg et al. \(2004\)](#);
- Article in journal: ([Fernandes et al., 2017](#)) and [Resenes et al. \(2019\)](#);
- Article in conferences proceedings: [Hölbjg et al. \(2014\)](#);
- Online documents or websites: ([Chang et al., 2019](#)) and [R Core Team \(2019\)](#);
- Masters/Thesis: [Nicolau \(2018\)](#);
- Technical reports: ([Hölbjg and Krämer, 2003](#)).

In your bib file **not use** `doi` and `url` fields. If the reference has DOI, use the `note` field to put the complete DOI address of this reference (e.g., `bibitem @article{Fernandes2017,...}` in the `references.bib` file). If the reference is online but does not have DOI, use the `note` field as follows: “Available at <http://xxx.xxx>” (e.g., `bibitem @Manual{R2019,...}` in the `references.bib` file).

For reference citations (using `\citep` or `\cite` commands), the citation should be linked to their respective reference, according to examples presented previously (task performed automatically by the RBCA $\LaTeX$ class).

2.2.1 This is a 3rd level heading

Use `\subsubsection` to get a 3rd level heading.

2.2.1.1 This is a 4th level heading.

Use `\paragraph` to get a 4th level heading. **Not use a 5th**

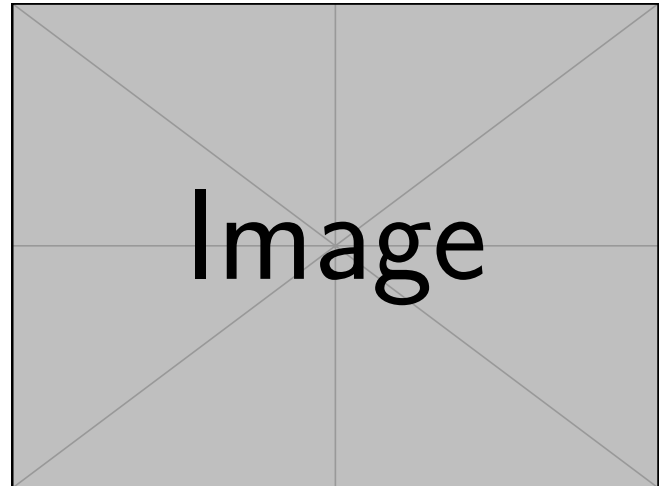


Figura 1: An example figure

level heading in your manuscript.

2.3 Figures and Tables

Figures and tables are added with the usual `figure` and `table` environments, e.g., [Figs. 1 e 2](#) and [Tabela 1](#). Use `figure*` and `table*` if you need a two-column wide figure or table, as in [Fig. 2](#) and [Tabela 2](#). If your table has a note, you can use `threeparttable` and `tablenotes` environments, as in [Tabela 1](#).

For the citation of figures, tables and equations use `\cref{}` command. Their numbers should be linked to their figures, tables, and equations (task performed automatically by the RBCA $\LaTeX$ class).

Tabela 1: Fonts and styles of the manuscript

Item	Font size	Font style
Title of paper	16 pt	bold
Authors	13 pt	bold
Institution	9 pt	–
Author e-mail	8 pt	–
PAPER CATEGORY	12 pt	bold
Text of Abstract/Resumo	9 pt	–
Text of paper	9 pt	–
Figure and Table caption	9 pt	–
1 Section	11 pt	bold
1.1 Subsection	10 pt	bold
1.1.1 3rd level	9 pt	bold
1.1.1.1 4rd level	9 pt	italic

This is a table note.
* Another note.

2.4 Formulas and equations

Equations and formulas should be placed on a new line, centralized and numbered consecutively for reference purposes, using the `equation` environment, as can be seen in

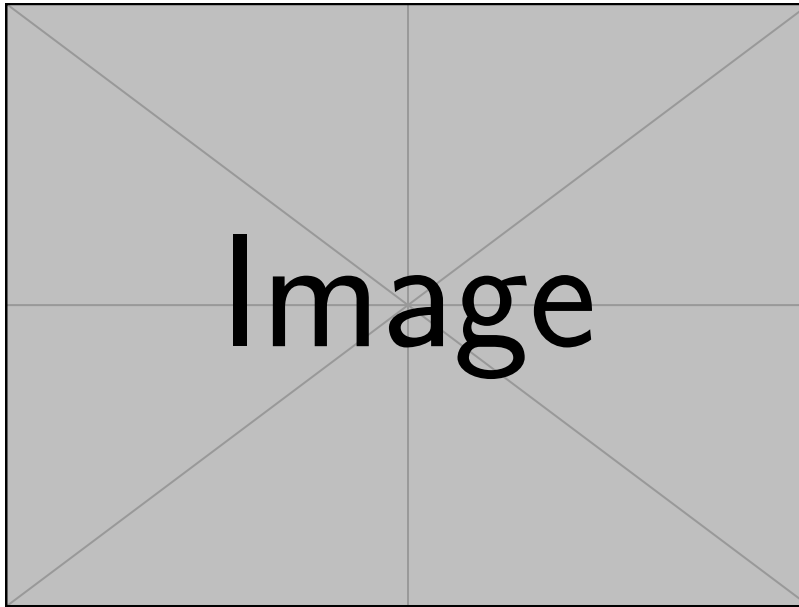


Figura 2: An example wide figure

Tabela 2: Matrix multiplication with *imatrix* data (adapted from (Hölbig and Krämer, 2003))

Program in:	Result/Matrix order	128 × 128	256 × 256	512 × 512	1024 × 1024
C-XSC	sequential	1.04666	5.11339	27.49690	204.10700
using long accumulator	8 processors	0.28522	1.17378	6.36642	45.96950
	speedup	3.66965	4.35634	4.31905	4.44005
C-XSC	sequential	0.63083	3.41555	21.16520	222.75400
with	8 processors	0.24378	1.16967	6.43267	43.80560
BLAS routines	speedup	2.58763	2.92009	3.29026	5.08505
C-XSC with	sequential	0.42283	1.69116	6.78124	31.38550
Dotk=1 and	8 processors	0.11418	0.44201	1.76142	8.72964
BLAS routines	speedup	3.70316	3.82604	3.84987	3.59527

Eqs. (1) e (2) or in Eq. (3).

$$\Delta T_{(ij)}(k) = T_{(ij)}(k) - T_{(ij)}^*(k) \quad (1)$$

$$T_{moc}(k) = T_{(ij)}(k) - \Delta T_{(ij)}(k) \quad (2)$$

$$BIAS = \frac{1}{M_{total}} \sum_{i=1}^{M_{total}} (T_i - T_i^*) \quad (3)$$

should be linked to their listing (task performed automatically by the RBCA $\LaTeX$ class).

```

1 require(dosNOW)
2 cl<-makeCluster(4) # number of cores
3 registerDoSNOW(cl)
4 # create a function check()
5 check <-function(n) {
6   for(i in 1:1000) {
7     sme <- matrix(rnorm(100), 10, 10)
8     solve(sme)
9   }
10 }
11 times <- 100 # times to run the loop
12 foreach(j=1:times) %dopar% check(j)
13 stopCluster(cl)

```

Figura 3: Source code in R language using `lstlisting` package

2.5 Program code listings

Program code listings should use the `lstlisting` package and are considered as figures. For reference purposes, we recommended that the lines of code be numbered. For example, the code of Fig. 3 shows a code in R language, where line 5 starts a function. The $\LaTeX$ code in the template shows how to set up a code listing. In the citation (using `\citet{}` command) of code listings, their number

2.6 Size, margins and footnotes of the manuscript

The page type used by RBCA is the “letter paper – letterpaper”, with a size defined as 19cm × 28cm. The left margin has 2.5cm, and the right margin is 2cm. The top margin has 2cm, and the bottom margin is 2.5cm. In [Tabela 1](#), we show the fonts and styles of the manuscript. Footnotes can be used throughout the text using `\footnote1` command. This formatting is performed automatically by the RBCA $\LaTeX$ class.

2.7 Abbreviations

The abbreviations used in the text are defined of this way: “word or text (abbreviation)”.

3 Submission of manuscripts

Submission of the manuscripts should follow the guidelines presented on the RBCA website. The website describes the entire evaluation process of manuscripts submitted to the Journal.

Acknowledgments

The Acknowledgments section should be placed at the end of the manuscript, before the References section, without numbering.

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¹Footnote example